

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of B. Tech. Examination (IT/CE/EC/CSE/EE/ME/CL)****April-May 2018****PY 141/ PY101 Engineering Physics****Date: 03.05.2018, Thursday****Time: 01:30 p.m. to 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of non-programmable scientific calculator is allowed.
5. Useful values of the constants are given on the last page.

SECTION – I**Q - 1 Choose the correct answer for the following questions. [07]****(a) Which of the following laser has four level? [01]**

- | | |
|-----------------------------|-------------------|
| (i) Diode laser | (ii) Nd:YAG laser |
| (iii) CO ₂ laser | (iv) He-Ne laser |

(b) The atomic weight of silicon is 28.0855. Round this value to 4 significant figures. [01]

- | | |
|-------------|------------|
| (i) 28.08 | (ii) 28.09 |
| (iii) 28.85 | (iv) 28.95 |

(c) Which of the following physical quantity is directly proportional to energy of musical sound wave? [01]

- | | |
|-------------------|---------------|
| (i) reverberation | (ii) area |
| (iii) intensity | (iv) velocity |

(d) Rate of spontaneous emission depends on [01]

- | | |
|---|---|
| (i) Number of atoms in ground state | (ii) Number of atoms in excited state |
| (iii) (i) and density of incident radiation | (iv) (ii) and density of incident radiation |

(e) The electrostriction phenomenon is useful for production of _____. [01]

- | | |
|----------------------|-----------------------|
| (i) Light | (ii) X- rays |
| (iii) electric field | (iv) ultrasonic waves |

(f) Two capacitors are connected parallel with each other. Find the maximum percentage error in the equivalent capacitance $\left(\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}\right)$ for the given [01] $C_1 = (10 \pm 0.3)\mu F$ and $C_2 = (20 \pm 0.4)\mu F$.

- | | |
|----------------|--------------|
| (i) 2.3 % | (ii) 0.023 % |
| (iii) 0.0023 % | (iv) 23 % |

(g) Total internal reflection within optical fibre wall can occur if the refractive index n_1 of the core must always be _____ the refractive index n_2 of the cladding. [01]

- | | |
|------------------|--------------------|
| (i) greater than | (ii) equal to |
| (iii) less than | (iv) None of these |

Q - 2 (a) The sound measured 100 m from a source is 120 dB. What is the sound level at 300 m? [04]**OR****Q - 2 (a) In ohm's experiment, the amount of current draw with different resistors were found to be 5.62 mA, 5.25 mA, 5.05 mA, 5.36 mA and 5.44 mA. Find out the average absolute error, relative error, percentage error and standard deviation. [04]**

- Q - 2 (b) Answer any two questions. [10]**
- (i) Explain construction and emission of 1064 nm radiation using energy level diagram of Nd:YAG laser.
 - (ii) Differentiate between Accuracy and Precision with example. Give classification of error in detail.
 - (iii) Describe the principle and method of producing ultrasonic waves using piezoelectric generator.

- Q - 3 (a) [04]**
- (1) What is Numerical Aperture (NA) and obtain an expression in terms of relative refractive index (Δ).
 - (2) If the velocity of light in core medium of an optical fibre is 1.94×10^8 m/s and refractive index of clad medium is 1.5, calculate acceptance angle and numerical aperture of the fibre.

OR

- Q - 3 (a) [04]**
- A volume of lecture room is 1000 m^3 and reverberation time of 2.0 s. If 100 students are increased in lecture room, what will be the new reverberation time of the lecture room? (The absorption of each student is 0.5 O.W.U.)

- Q - 3 (b) Answer any two questions. [10]**
- (i) Write down factors affecting acoustics of buildings and their remedies.
 - (ii) Define reverberation time and state its importance. Explain the measurement of absorption coefficient using reverberation time.
 - (iii) Write down primary advantages of fiber optic communication compared to metallic cable communication.

SECTION – II

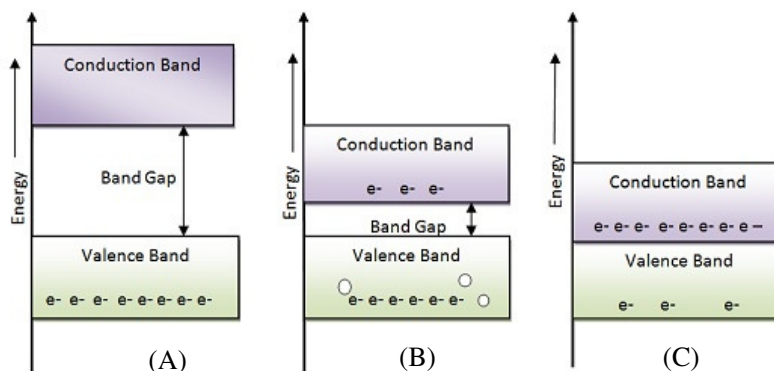
- Q - 4 Choose the correct answer for the following questions. [07]**

- (a) Positive hall coefficient (R_H) refers to _____. [01]**
- | | |
|-----------------------|-----------------------|
| (i) neutral material | (ii) p-type material |
| (iii) n-type material | (iv) pn-type material |
- (b) The Lattice parameter of monoclinic crystal system is _____. [01]**
- | | |
|---|--|
| (i) $a = b \neq c$ and $\alpha = \beta = \gamma = 90^\circ$ | (ii) $a = b = c$ and $\alpha = \beta = \gamma = 90^\circ$ |
| (iii) $a \neq b \neq c$ and $\alpha = \beta = \gamma \neq 90^\circ$ | (iv) $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90^\circ$ |
- (c) CNT is _____ and Bucky ball is _____ type of nanomaterial. [01]**
- | | |
|--------------|-------------|
| (i) 0D, 1D | (ii) 2D, 3D |
| (iii) 1D, 0D | (iv) 1D, 2D |
- (d) Which of the following property is false for X-rays? [01]**
- | | |
|---------------------------------------|--|
| (i) X-rays have no mass | (ii) X-rays are invisible |
| (iii) X-rays can be optically focused | (iv) wavelength range of X-rays is $0.1 - 100 \text{ \AA}$ |
- (e) The Coordination number of BCC structure is _____. [01]**
- | | |
|---------|---------|
| (i) 8 | (ii) 12 |
| (iii) 6 | (iv) 4 |

(f) Which of the following method can be used for optical characterization of nano material? [01]

- (i) Transmission Electron Microscopy (ii) X-ray diffraction
(iii) Photoluminance (iv) EDS

(g) Identify the type of material based on band structure in the following figure. [01]



- (i) (A) insulator (B) semiconductor (C) conductor
(ii) (A) semiconductor (B) insulator (C) conductor
(iii) (A) insulator (B) conductor (C) semiconductor
(iv) (A) conductor (B) insulator (C) semiconductor

Q – 5 (a) Write four application of Hall effect. [04]

The Hall coefficient of germanium was found to be $-5.25 \times 10^{-5} \text{ m}^2 \text{C}^{-1}$. Determine the nature of semiconductor if the conductivity was found to be $200 \Omega^{-1} \text{m}^{-1}$. Calculate the density and mobility of the charge carrier.

OR

Q – 5 (a) Write procedure to find the Miller Indices. [04]

Q – 5 (b) Answer any two questions. [10]

- (i) Classify solid materials using band theory.
(ii) Describe the following methods to synthesize nano particles.
(A) CVD (B) Ball milling.
(iii) (1) Write any six characteristics of X-rays
(2) Differentiate crystalline solid and amorphous solid

Q – 6 (a) Draw the neat diagrams of following planes: $(\bar{1}01)$, (110) , (210) and $(00\bar{1})$ [04]

OR

Q – 6 (a) Define the following terms (Use suitable examples/ diagrams if required): [04]
(1) Basis (2) Meissner effect (3) Isotope effect (4) Unit cell

Q – 6 (b) Answer any two questions. [10]

- (i) Differentiate soft and hard superconductors.
For mercury of mass number 202, the value α is 0.70 and T_c is 4.22 K. Find the transition temperature for the isotope of mercury of mass number 200.
(ii) What is SEM? Explain construction and working of SEM.
(iii) (A) FCC crystal has atomic radius $1.246 \times 10^{-10} \text{ m}$. Calculate the interplanar spacing between (101) and (220) planes in crystal.
(B) Define co-ordination number and atomic packing fraction.

Table 1		
Sr. no.	Name of constant	Value of constant
1	Speed of light (c)	$3 \times 10^8 \text{ m/s}$
2	Planck's Constant (h)	$6.626 \times 10^{-34} \text{ Js}$
3	Boltzmann Constant (k)	$1.38 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$
4	Electric Charge (e)	$1.6 \times 10^{-19} \text{ C}$
5	Avogadro Number (N_A)	$6.022 \times 10^{23} \text{ mol}^{-1}$

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